Extented Generalised Pareto Distribution (EGP)

Parametrisation

The cummulative distribution function for the EGP distribution, is

$$F(y) = \left(1 - \left(1 + \xi \frac{y}{\sigma}\right)^{-1/\xi}\right)^{\kappa}, \qquad y > 0.$$

Here,

- ξ is the tail-parameter,
- $\sigma > 0$ is the scale-parameter, and
- $\kappa > 0$ is the shape-parameter.

Link-function

The linear predictor η controls the α-quantile of EGP

$$\text{Prob}(y \leq q_\alpha) = \alpha.$$

σ can be expressed in terms of the quantile

$$\sigma = \frac{\xi q_\alpha}{(1 - \alpha^{1/\kappa})^{-\xi} - 1}$$

and the quantile is linked to the linear predictor as

$$q_\alpha = \exp(\eta).$$

Hyperparameters

There are two hyperparameters, the tail and shape parameter, represented as (θ_1, θ_2). The tail-parameter is in a predefined range (ξ_L, ξ_H)

$$\xi = \xi_L + (\xi_H - \xi_L) \frac{1}{1 + \exp(-\theta_1)}$$

and the prior is defined on θ_1. The shape-parameter is

$$\kappa = \exp(\theta_2)$$

and the prior is defined on θ_2.

Specification

- `family="egp"`
- Required arguments: the observations y
- The quantile level α

Hyperparameter spesification and default values

doc Exteneded Generalized Pareto likelihood

hyper

theta1

hyperid 101211
name tail
short.name xi
output.name Tail parameter for egp observations
output.name.intern Intern tail parameter for egp observations
initial 0
fixed FALSE
prior pc.egptail
param 5 -0.5 0.5
to.theta function(x, interval = c(REPLACE.ME.low, REPLACE.ME.high)) log(-(interval[1] - x) /
from.theta function(x, interval = c(REPLACE.ME.low, REPLACE.ME.high)) interval[1] + (interval

theta2

hyperid 101212
name shape
short.name kappa
output.name Shape parameter for the egp observations
output.name.intern Intern shape parameter for the egp observations
initial 0
fixed FALSE
prior loggamma
param 100 100
to.theta function(x) log(x)
from.theta function(x) exp(x)

survival FALSE

discrete FALSE

link default quantile

pdf egp

Example

```
n <- 3000
for (xi in c(-0.3, 0.3)) {
    xi.intern <- inla.models()$likelihood$egp$hyper$theta1$to.theta(xi, interval=c(-0.5, 0.5))
    alpha <- 0.9
    kappa <- 1.1
    kappa.intern <- inla.models()$likelihood$egp$hyper$theta2$to.theta(kappa)
    a <- ((1-alpha^(1/kappa))^(-xi) -1)
    x <- rnorm(n, sd = 0.3)
    eta <- 0.9 + 1.1 * x
    q <- exp(eta)
    sigma <- xi * q / a
    y <- - (sigma / xi) * (1- (1-runif(n)^(1/kappa))^(-xi))

    r <- inla(y ~ 1 + x,
              family = "egp",
              control.family = list(
                  control.link = list(quantile = alpha),
                  hyper = list(tail = list(
                                   ##initial = xi.intern,
                                   fixed = !TRUE,
                                   prior = "pc.egptail",
                                   param = c(5, -0.5, 0.5)),
                               shape = list(
                                   ##initial = kappa.intern,
                                   fixed = !TRUE,
                                   prior = "loggamma",
                                   param = c(100, 100)))),
              data = data.frame(y, x),
              verbose = !TRUE)
    print(summary(r))
}
```